Federated Learning with Secure Aggregation for Privacy-Preserving Deep Learning in IoT Environments

Vinay Kumar Kasula
Department of Information Technology, University of the Cumberlands,
Williamsburg, KY, USA
vinaykasula.phd@ieee.org

Mounica Yenugula
Department of Information Technology, University of the Cumberlands,
Williamsburg, KY, USA
ymounica.phd@ieee.org

Bhargavi Konda
Department of Information Technology, University of the Cumberlands,
Williamsburg, KY, USA
bhargavikonda@ieee.org

Akhila Reddy Yadulla
Department of Information Technology, University of the Cumberlands,
Williamsburg, KY, USA
akhilareddyyadulla@ieee.org

Chaitanya Tumma
Department of Information Technology, University of the Cumberlands,
Williamsburg, KY, USA
ctumma30994@ucumberlands.edu

Sarath Babu Rakki
Department of CSE
JNTU Hyderabad
Hyderabad, India
sharath.rakki@gmail.com

***Abstract*— To effectively handle the massive data generated by large-scale IoT applications, deep learning has been widely applied in IoT environments. However, deep models face security threats such as inference attacks and model reverse-engineering during the training process, which can lead to the leakage of original data fed into the model. Federated Learning (FL) with Secure Aggregation offers a robust solution to this issue by enabling distributed model training without sharing raw data. This method ensures privacy by securely aggregating model updates from IoT devices, mitigating risks of data exposure during transmission. To further enhance model performance, a regularization term is introduced to address overfitting and facilitate the learning of significant features. Experimental results demonstrate that the proposed approach effectively enhances the generalization capability of the model. As the number of iterations increases, the accuracy difference between the FL-trained model and the model trained on centralized data remains below 0.5%. Therefore, the proposed method not only protects user privacy but also ensures model usability, achieving a balance between privacy and utility.**

Keywords*— *IoT, Federated Learning, Secure Aggregation, regularization, deep learning, privacy preservation.

I. Introduction

In recent years, IoT technology has rapidly developed, finding widespread applications in various domains such as wearable devices, environmental monitoring, logistics management, virtual reality, smart homes, smart grids, smart cities, and public services. This has closely connected the digital and physical worlds, accelerating the advent of the Internet of Everything [1]. With the large-scale deployment of IoT, massive data involving personal privacy, such as electricity usage, health information, and movement trajectories, is being recorded and collected by IoT devices [2]. To efficiently handle the vast amounts of data generated by IoT applications, deep learning is widely used in model training within IoT environments [3]. However, this also introduces potential risks to privacy. During the training process, model parameters may face security threats such as inference attacks and model reverse-engineering, which could lead to the leakage of original data input into the model [4]. To address this issue, protection of the model's training parameters is essential. Common privacy protection techniques for deep learning include homomorphic encryption and differential privacy. Homomorphic encryption does not impact model accuracy but introduces ciphertext expansion, resulting in significant communication overhead, which makes it less suitable for IoT environments [5]. On the other hand, differential privacy can protect model parameters by adding noise to the training data, which may result in a slight loss of accuracy. Balancing model accuracy and privacy protection is a key concern [6].
In this paper, we adopt Federated Learning (FL) with Secure Aggregation as an alternative solution to ensure data privacy during the model training process. Federated learning enables decentralized model training without directly sharing sensitive data, thus ensuring user privacy [7]. Secure aggregation ensures that only the final aggregated model parameters are shared, maintaining data confidentiality [8]. This approach not only protects user privacy but also ensures the usability of the trained models by combining privacy preservation with model accuracy [9]. Through this approach, we aim to provide a framework for securely training deep learning models in IoT environments, achieving a balance between privacy and usability without the overhead of traditional encryption methods.

II. Privacy Protection Techniques and Optimization Models

A. Federated Learning with Secure Aggregation

Federated Learning (FL) is a distributed machine learning technique where models are trained collaboratively across multiple decentralized devices, without sharing raw data. In FL with Secure Aggregation, privacy is enhanced by ensuring that individual device data remains confidential, even from the central server. The key idea is that, instead of sharing raw data, devices share model updates, which are aggregated securely to protect user privacy.
Secure Aggregation in Federated Learning: In FL-SA, each device D_i trains its model locally based on its private data and then shares the model update with the central server. The server aggregates these updates without learning about individual contributions, preserving privacy. Let f_i represent the model update from device D_i, which could be a vector of

gradients or weights. The server aggregates these updates securely. We ensure that the aggregated model update is indistinguishable from the updates coming from neighboring datasets $D_1$and D_2.

Mathematically, this can be expressed as:

$\Pr(\text{Aggregate}(f_1) \in S) \leq \exp(\epsilon) \cdot \Pr(\text{Aggregate}(f_2) \in S)$

where $\text{Aggregate}(f_1)$ and $\text{Aggregate}(f_2)$ are the aggregated updates from devices D_1 and D_2, respectively, ϵ is the privacy budget, controlling how much noise is added to the aggregated updates to protect privacy. A smaller ϵ provides stronger privacy guarantees. S represents a subset of possible aggregated updates.

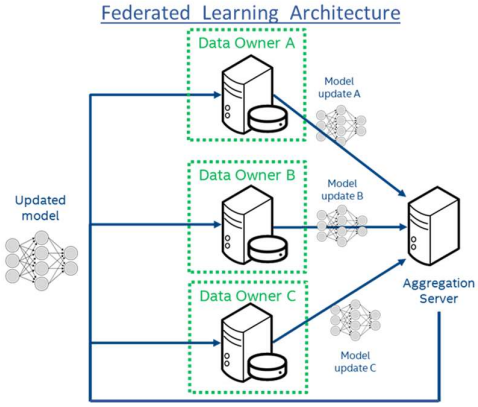


FIGURE 1. STEPS OF ADAPTIVE GRADIENT NOISE INJECTION

The key difference here is that, instead of adding noise to individual queries (as in DP), noise is added to the aggregated model update to protect the privacy of each device's contribution to the global model.

Secure Aggregation Sensitivity: In FL-SA, the sensitivity of the model updates is crucial for determining how much noise needs to be added. The sensitivity of the function f (the model update computation) measures the maximum difference between the model updates from any two neighboring datasets.

Let $f: D \to R^d$ be the function that computes the model update on dataset D, where D represents a local dataset at device D_i. The sensitivity Δf is defined as the maximum difference between the updates $f(D_1)$ and $f(D_2)$ for any two neighboring datasets D_1 and D_2:

$$\Delta f = \max_{D_1, D_2} \| f(D_1) - f(D_2) \|_1$$

where $\| f(D_1) - f(D_2) \|_1$ is the L_1-norm distance between the updates computed for the two neighboring datasets D_1 and D_2. f(D) is the model update for dataset D. The smaller the sensitivity Δf, the less noise needs to be added to ensure privacy during aggregation.

Laplace Mechanism for Secure Aggregation: To preserve privacy during aggregation, noise is often added using a Laplace distribution. In FL-SA, the Laplace mechanism ensures that the aggregated updates remain indistinguishable from each other, even when combined with updates from neighboring datasets.

The Laplace distribution has a mean of 0 and scale parameter b, denoted as Lap(b), with the probability density function:

$$p(x) = \frac{1}{2b} exp\left(-\frac{|x|}{b}\right)$$

The aggregated model update f_{agg} after adding Laplace noise is given by:

$$fagg = \sum_{i=1}^{n} f_i + \text{Lap}\left(\frac{\Delta f}{\epsilon}\right)$$

where f_i is the update from device D_i (local model updates), $\text{Lap}\left(\frac{\Delta f}{\epsilon}\right)$ is the Laplace noise added to the aggregation to ensure ϵ\epsilonϵ-privacy, Δf is the global sensitivity, and ϵ is the privacy budget.

This equation ensures that the aggregated update is protected by the Laplace noise, providing privacy for the individual updates.

Sequential Composition for Secure Aggregation: In federated learning, when there are multiple rounds of aggregation, the privacy budget for each round adds up. If each round K_i satisfies ϵ_i -privacy, then the overall privacy guarantee is the sum of the individual privacy budgets:

$$\sum_{i}^{n} \epsilon_i - privacy$$

This means that for each additional round of aggregation, the privacy budget accumulates. Therefore, if there are multiple rounds of model updates, the total privacy budget used across all rounds is the sum of the privacy budgets for each round.

B. Regularization

Regularization in federated learning addresses the problem of overfitting by introducing a penalty on the complexity of the model. It encourages simpler models, ensuring that local models do not overfit to individual datasets, which can be highly heterogeneous across devices. The regularization term ensures that model parameters (weights) remain small and the model generalizes well across all devices. The regularization objective function can be written as:

$$J(\omega) = \frac{1}{2m}\sum_{i=1}^{m}[y_i - h(x_i)]^2 + \lambda \sum_{j=1}^{n} \omega_j^2$$

Where m is the number of samples in the dataset, n is the number of features, λ is the regularization hyperparameter that controls the strength of the regularization, ω_j represents the weights (parameters) of the model, y_i is the target value for sample i, $h(x_i)$ is the predicted output from the model for input x_i.

The regularization term $\lambda \sum_{j=1}^{n} \omega_j^2$ prevents overfitting by penalizing large weights, ensuring that the model learns generalizable patterns across devices.

C. Neural Networks

Neural networks in FL are trained using Stochastic Gradient Descent (SGD) or its variants, and the updates are aggregated across devices. The training process involves minimizing the loss function by adjusting model parameters based on gradients.

The update rule in federated learning can be expressed as:

$$\omega_j = \omega_j - \alpha \frac{\partial J(\omega)}{\partial \omega_j}$$

Where α is the learning rate (step size), $\frac{\partial J(\omega)}{\partial \omega_j}$ is the gradient of the loss function with respect to the weight vector ω_j, The weights ω_j are updated by moving in the direction of the negative gradient to minimize the loss function.

In FL, these updates are securely aggregated to form the global model, preserving the privacy of individual updates from each device.

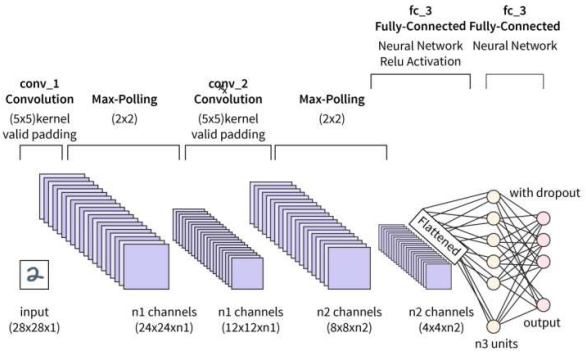


FIGURE 2. NEURAL NETWORK ARCHITECTURE AND PARAMETERS

III. PRIVACY BUDGET ALLOCATION SCHEME

In Federated Learning (FL), parameter updates during training are nonlinear and non-uniform, typically decreasing as the model converges. This section introduces a scheme for allocating privacy budgets during each training round while maintaining differential privacy (DP) principles within the secure aggregation framework. This ensures that noise interference is minimized under a given privacy budget, leading to faster model convergence. Additionally, a regularization term is introduced to limit noise and mitigate overfitting. The adaptive gradient noise addition process in FL with Secure Aggregation is conceptually outlined below.

In FL, clients collaboratively train a global model without sharing raw data, instead exchanging encrypted gradient updates via Secure Aggregation. This prevents intermediate or malicious servers from learning individual client updates. To enhance privacy, noise is added to gradients before encryption, with varying noise levels determined by allocated privacy budgets for each round.

A. Privacy Budget Allocation

Traditional privacy-preserving FL allocates a fixed privacy budget for each training round, injecting the same amount of noise. However, during early rounds, when gradients are larger, small inaccuracies do not significantly affect optimization. As the model converges and gradients shrink, precise gradient measurements are crucial. A fixed allocation in these later stages can introduce excessive noise, degrading convergence. This scheme instead allocates privacy budgets dynamically.

Using a linear increment approach, the privacy budget for the rth training round is increased by a fixed value Δ_b ($\Delta_b \geq 0$), resulting in the privacy budget for the rth round:

$$\beta_r = \beta_1 + (r-1)\Delta_b, 1 \leq r \leq R$$

Where β is the privacy budget for the rth round, R is the total number of training rounds, Δ_b\ is the fixed increment per round.

The total privacy budget across all R rounds is:

$$\beta_{total} = \beta_1 + \beta_2 + \cdots + \beta_R$$

Alternatively, this can be expressed as:

$$\beta_{total} = R\beta_1 + \frac{R(R-1)}{2}\Delta_b$$

By combining Equations (2) and (3), the total privacy budget becomes:

$$\beta_{total} = R\left(\beta_1 + \frac{R-1}{2}\Delta_b\right)$$

From this, the initial privacy budget β_1 can be derived:

$$\beta_1 = \frac{\left(\beta_{total} - \frac{R(R-1)}{2}\Delta_b\right)}{R}$$

Substituting Equation (5) into Equation (1), the privacy budget for the rth round is:

$$\beta_r = \frac{\beta_{total}}{R} + (r-1)\left(\Delta_b - \frac{\beta_{total}}{R}\right)$$

To ensure non-negative budgets and increments:

$$0 \leq \Delta_b \leq \frac{2\beta_{total}}{R(R-1)}$$

Given the total privacy budget, this formulation provides the privacy budget for each training round.

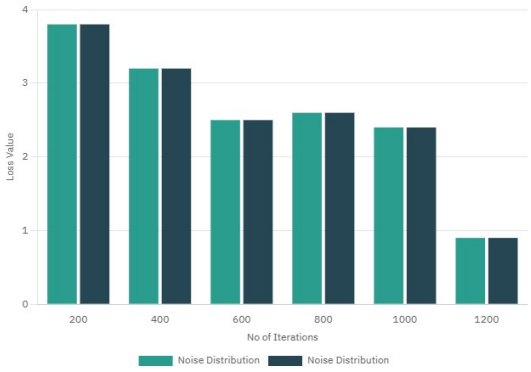


FIGURE 3. COMPARISON OF LOSS FUNCTION CHANGES ACROSS ITERATIONS

B. Noise Value Constraints in Secure Aggregation

In FL, dynamic privacy budget allocation with fixed increments Δ_b may result in smaller budgets for some rounds, leading to larger noise. To mitigate the impact of noise, the noise value is constrained within a certain range. In Secure Aggregation, noise is added locally by clients before encryption. This approach uses regularization to limit noise and prevent overfitting.

L2 Regularization is employed to constrain the aggregated gradient's magnitude. The regularization term is added to the global loss function:

$$Regularization\ Term: \lambda \sum_{i=1}^{n} \theta_i^2$$

where λ is the regularization coefficient, n is the number of parameters, θ_i represents the global model parameters.

The global loss function, incorporating the regularization term, becomes:

$$L = L_0 + \frac{\lambda}{n}\sum_{i=1}^{n} \theta_i^2$$

where L_0 is the original loss function. The gradient of the regularized loss is:

$$\frac{\partial L}{\partial \theta} = \frac{\partial L_0}{\partial \theta} + \frac{\lambda}{n}\theta$$

When updating the model parameters:

$$\theta_{r+1} = \theta_r - \eta\left(\frac{\partial L_0}{\partial \theta} + \frac{\lambda}{n}\theta\right)$$

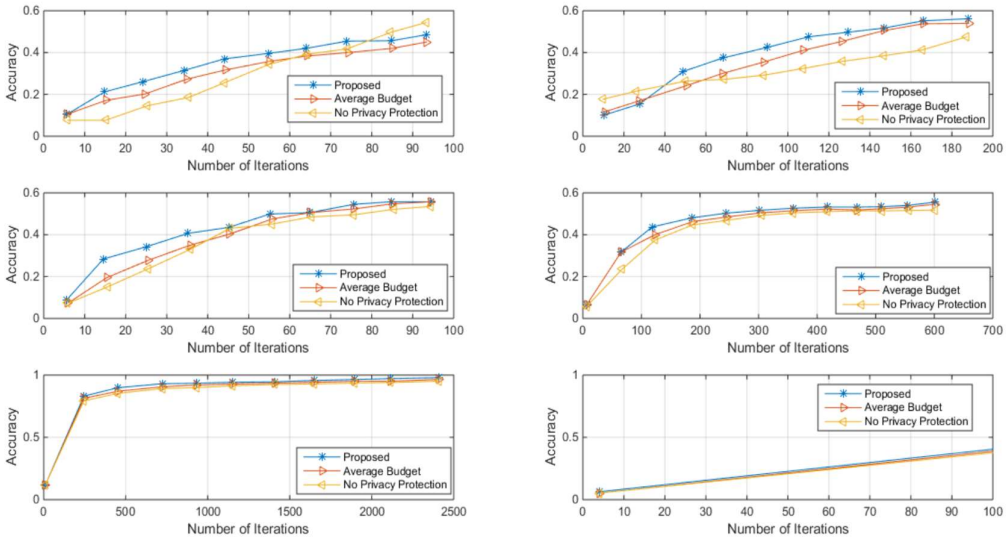


FIGURE 4. COMPARISON OF ACCURACY CHANGES AT MODEL ITERATIONS 100, 200, 300, 600, 2,400, AND 3,600

For gradients with added noise under Secure Aggregation:

$$\theta_{r+1} = \theta_r - \eta\left(\frac{\partial L_0}{\partial \theta} + N(0,\sigma^2)\right)$$

θ_{r+1} updated parameter at iteration $r+1$, θ current parameter at iteration r, η learning rate, $\frac{\partial L_0}{\partial \theta}$ gradient of the loss function L_0 with respect to θ. $N(0,\sigma^2)$ gaussian noise added to the gradient for differential privacy, where σ^2 controls the noise scale.

The noise term is compared with the regularization term. If the noise is smaller than the regularization term, the standard gradient descent process continues. Otherwise, the noise term is replaced with the regularization term for finer gradient processing. By applying regularization, model weights are reduced, ensuring that small input variations do not significantly alter model behavior. This prevents overfitting to noisy data and helps the model focus on essential patterns in aggregated gradients. If regularization is used, the privacy budget consumption may need recalibration.

C. Privacy Budget Reallocation in Federated Learning with Secure Aggregation

In Federated Learning (FL) with Secure Aggregation, noise addition and budget allocation are optimized to ensure that total privacy budget consumption remains consistent across iterations. This section outlines how to calculate and redistribute privacy budgets effectively while maintaining model performance and security. Recalculation of Consumed and Remaining Privacy Budgets

To account for the privacy budget consumed by the secure aggregation mechanism in the tth iteration, the following formulation applies:

$$\epsilon_t' = \frac{\Delta g \cdot \lambda}{m \cdot \epsilon_t}$$

where ϵ_t' privacy budget consumed in the tth iteration, Δg sensitivity of the aggregated gradient, m number of participants.

The remaining privacy budget after t iterations is computed as:

$$\epsilon' = \epsilon - \sum_{i=1}^{t} \epsilon_i' \Rightarrow \epsilon' = \epsilon - \epsilon_t' - \sum_{i=1}^{t-1} \epsilon_i'$$

For the next iteration, the available privacy budget is dynamically adjusted based on the remaining budget:

$$\epsilon_{t+1} = \frac{\epsilon' - (T-t-1)\cdot \Delta_b}{T-t}$$

This process repeats until the total training rounds T are completed, or the model converges.

Algorithm: Privacy Budget Allocation in Federated Learning

The detailed process is outlined in Algorithm 1, which adapts privacy budgets dynamically while ensuring robust training through secure aggregation.

Algorithm 1: Dynamic Privacy Budget Allocation in FL with Secure Aggregation

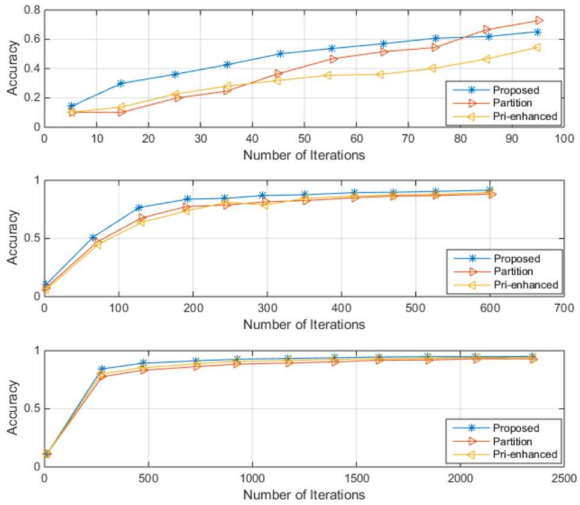


FIGURE 5. THE IMPACT OF DIFFERENT ITERATION COUNTS ON ACCURACY WITH FIXED PRIVACY BUDGETS

Input: Total privacy budget ϵ, Gradient aggregation sensitivity Δg, Incremental privacy budget Δ_b, Regularization coefficient λ.
Output: Final model weights ω.
Steps:

- Initialize the model weights ω_t.
- For each round $t = 1,2, \dots, T$:
  - Compute allocated privacy budget for the round:
    $$\epsilon_t = \frac{\epsilon}{T} + \frac{t \cdot \Delta_b}{2}$$
  - Determine the noise magnitude:
    $$\Delta Lap = \frac{\Delta g}{m \cdot \epsilon_t}$$
  - Calculate the regularization term value:
    $$r_t = \lambda \cdot \omega$$
  - Conditionally update the weights:
    - Case 1: If the added noise is smaller than or equal to the regularization term:
      $$\omega_{t+1} = \omega_t - \eta(\nabla L(\omega_t) + \Delta Lap)$$
      Update the privacy budget for the next round:
      $$\epsilon_{t+1} = \epsilon_t + \Delta_b$$
    - Case 2: If the noise exceeds the regularization term: Use the regularization term to limit noise addition:
      $$\omega_{t+1} = \omega_t - \eta(\nabla L(\omega_t) + r_t)$$
      Calculate the privacy budget consumed by the regularization term:
      $$\epsilon_t' = \frac{\Delta g \cdot \lambda}{m \cdot \epsilon_t}$$
      Adjust the remaining privacy budget:
      $$\epsilon' = \epsilon - \sum_{i=1}^{t} \epsilon_i'$$
  - Repeat until all iterations T are completed or convergence is reached.

end for
Return: Final model weights ω_T.

IV. Experiments and Analysis

A. Experimental Setup

To assess the effectiveness of the gradient noise injection approach, experiments were carried out using the Tiny ImageNet and DeepFashion datasets:

- Tiny ImageNet Dataset: This dataset consists of 64×64 color images across 200 classes, with 500 images per class. It includes 100,000 training images and 10,000 testing images, divided into 200 training batches and one testing batch, with equal distribution across batches. Tiny ImageNet is chosen for its increased complexity and greater variety compared to the CIFAR-100 dataset, providing a more challenging classification task.
- DeepFashion Dataset: This dataset includes over 800,000 images of fashion items, categorized into various attributes (e.g., type, color, style, etc.). DeepFashion offers a more comprehensive and real-world fashion dataset compared to Fashion-MNIST, making it suitable for testing models with more sophisticated label and feature identification.

For training, a popular deep learning framework was utilized to evaluate the gradient parameter variations during model iterations and their relationship with privacy budgets. The framework is designed to be efficient for IoT sensor nodes with limited resources. Hyperparameter settings followed the methodology of Abadi et al. [15], with the neural network architecture and parameters illustrated in Figure 2.

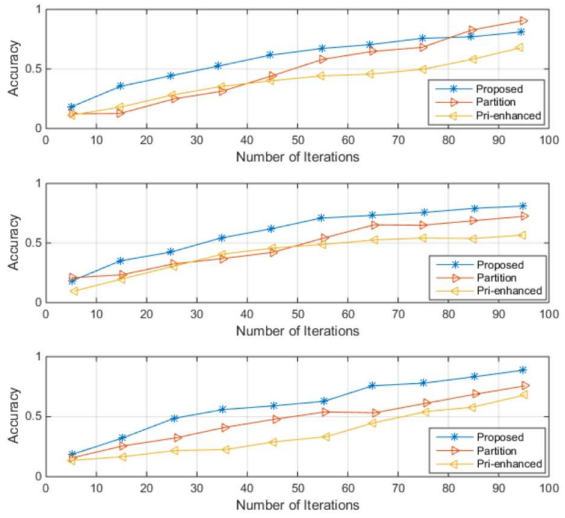


FIGURE 6. THE IMPACT OF DIFFERENT PRIVACY BUDGETS ON MODEL ACCURACY WITH 100 ITERATIONS

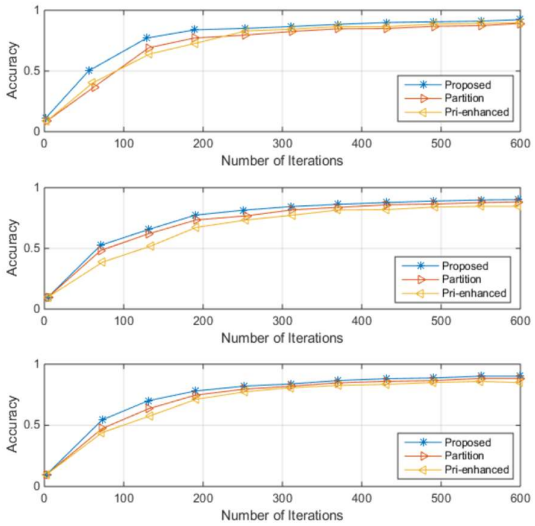


FIGURE 7. THE IMPACT OF DIFFERENT PRIVACY BUDGETS ON MODEL ACCURACY WITH 600 ITERATIONS

B. Experimental Analysis

Evaluation Metric: Model loss was used as the primary evaluation metric for assessing performance.

- Tiny ImageNet Experiments: The deep model was trained on the Tiny ImageNet dataset with random per-

turbations added to gradient parameters. The loss changes were recorded every 100 iterations, both before and after perturbation. The results indicated that introducing noise significantly enhanced the model's generalization ability, with the model achieving faster convergence in loss reduction compared to the unperturbed data. This demonstrated that appropriate noise helps the model learn more important parameters. Figure 3 compares the loss function's change over iterations.

- DeepFashion Validation: The method was fully tested on the DeepFashion dataset and compared with the equal privacy budget allocation strategy and a baseline without privacy protection. In differential privacy stochastic gradient descent, the parameters must be optimized, as they greatly influence model performance but are not directly learned.
- The experiment tested model accuracy across various iterations at fixed privacy budgets. For strong privacy protection, the budget was set at 0.5, with clipping set to 0.001, a regularization coefficient of 0.3, and a learning rate of 5×10^{-3}. Accuracy comparisons at iteration counts of 50, 150, 250, 500, 1,500, and 2,500 are shown in Figure 4. Early iterations revealed superior performance with the proposed method, as noise worked as a regularization term, pushing the model toward optimal solutions.
- Comparison with Other Methods: The results were compared with Gong et al.'s "pri-enhanced" method [24] and Wang's "partition" method [25]. The "pri-enhanced" method incrementally increases the privacy budget from minimum to maximum, while the "partition" method evenly distributes the budget across iterations.
- Experiments with fixed privacy budgets of 0.5, 2.5, and 5 were conducted under varying iteration counts. As shown in Figure 5, the proposed method consistently outperformed the baselines, especially in IoT scenarios with fewer iterations. At 50 iterations, the proposed method achieved higher accuracy compared to the baseline models.
- Effect of Varying Privacy Budgets: The impact of varying privacy budgets (0.5, 2.5, and 5) on model accuracy was tested with fixed iterations (e.g., 500). The results in Figure 7 confirm that the proposed method maintained its superiority, providing strong privacy protection without significant performance loss.

Performance Insights

- Fewer Iterations: The proposed method showed strong performance with fewer iterations, demonstrating the advantage of injecting noise to improve generalization. This method helps reduce unnecessary parameter adjustments, making the model less sensitive to "noisy" features, thus achieving high accuracy with fewer iterations a key benefit for resource-constrained IoT devices.
- Higher Iterations: As iteration counts increased, the reduced privacy budget per iteration resulted in more noise being introduced, slightly diminishing the relative advantage of the proposed method. However, the model's accuracy continued to improve, eventually approaching the performance of unperturbed training. This highlighted the balance between maintaining privacy protection and model utility, showing that the noise injection approach helps maintain robust performance even as privacy protection increases.

V. Conclusion

The proposed gradient noise injection approach demonstrated significant improvements in model performance across both the Tiny ImageNet and DeepFashion datasets. By introducing noise, the method enhanced the model's generalization ability, achieving faster loss convergence and better overall accuracy compared to unperturbed data. It performed particularly well with fewer iterations, making it advantageous for resource-constrained IoT devices. The method provided robust privacy protection without substantial performance degradation, even when the privacy budget was reduced. When compared to existing methods like "pri-enhanced" and "partition", the proposed approach outperformed them in scenarios with limited iterations. Overall, the results highlighted the balance between privacy and model utility, offering an effective solution for privacy-preserving machine learning.

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